REPORT

RESOLVE
MINING SOLUTIONS

Rio Tinto Oyu Tolgoi Airblast Modelling – P2N Early cHR Caving - FINAL

PROJECT REPORT

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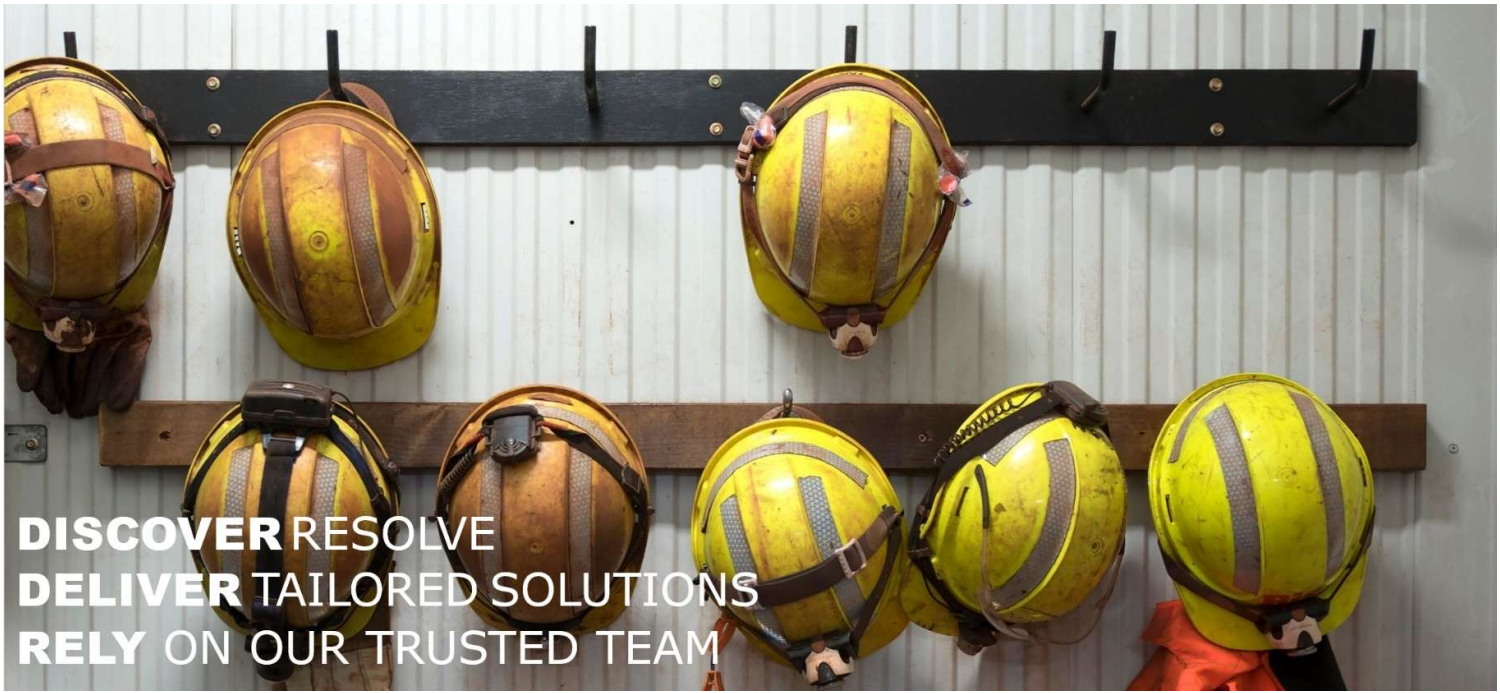

Report Prepared For

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EXECUTIVE SUMMARY

Rio Tinto – Copper Technical engaged Resolve Mining Solutions to assess the potential airblast hazard associated with the early propagation of the Panel 2 North (P2N) cave at the Hugo North deposit, Oyu Tolgoi. This study focused on the Critical Hydraulic Radius (cHR) phase, where sudden cave failure may occur without sufficient muckpile resistance, potentially leading to hazardous airblast events.

Using the VentSim™ Caving Airblast Simulation Tool (CAST), a series of scenarios were modelled to evaluate the risk posed by varying cave geometries, muckpile heights, and airgap volumes. The modelling revealed that airblast velocities can exceed safe operational thresholds (>20 m/s) when muckpile levels fall below the apex level and/or large airgaps form. Conversely, scenarios with muckpile cover above the apex level demonstrated significantly reduced airblast velocities, indicating effective mitigation.

Key findings include:

- **Current controls are suitable:** The planned muckpile controls currently in use at site are assessed to be appropriate for managing airblast risk. Ongoing monitoring of muckpile condition, supported by timely remediation where required, should remain a priority to ensure these controls remain effective during early cave propagation.
- **Infrastructure Vulnerability**: Apex and undercut drives are primary pathways for airblast propagation, with potential failure of access doors and ventilation infrastructure under high-pressure scenarios.
- **Airgap Sensitivity**: Airblast severity is highly sensitive to airgap volume and muckpile depletion, especially during early cave propagation.
- **Mitigation Effectiveness**: Engineering controls such as airblast plugs and ventilation circuit changes are limited in effectiveness due to the cave's central location. Operational controls like void management and access restrictions are more viable.

Proactive void and muckpile management coupled with access control are critical to mitigating airblast risk during the early stages of cave propagation at P2N. Recommendations include maintaining high muckpile levels, implementing or maintaining real-time monitoring (mass balance and direct monitoring systems), updating risk assessments and TARPs, and continuing scenario modelling to inform operational decisions.

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1 INTRODUCTION

Rio Tinto – Copper Technical (the client) engaged Resolve Mining Solutions (RMS) to assess the risk of early-stage critical hydraulic radius (cHR) airblast to the Panel 2 North (P2N) of the Hugo North cave at Oyu Tolgoi (OT). This report details the findings based on the agreed scope of work outlined in the section below.

2 SCOPE OF WORK

The Client engaged RMS to complete early stage cHR propagation airblast modelling of the P2N mine.

The Scope, is understood to be:

- Project meetings and collaboration with Client.
- Collating of data and input parameters.
- Review current VentSim™ mine model.
- Build and run the airblast model in VentSim™.
    - Base case cHR model with current mining and geotechnical assumptions.
    - Sensitivity of muckpile heights and airgaps for pre-cHR geometry.
- Risk assessment and mitigation recommendations.
    - Identify high velocity (>20m/s) areas of the mine due to airblast.
    - Provide conceptual high-level review of controls to mitigate the hazard.

3 DATA REVIEW

Table 3-1 summarises the data provided by the client to carry out the agreed scope of work.

Table 3-1 – Summary of files provided by the client

	FILE TYPE	FILE NAME(S)	COMMENTS
ENGINEERING	VentSim™ Model	• OT_VentSim_5Q3.Rev3b Model.vsm	*Site staged VentSim™ model.*
	Deswik File	• P2N Airblast Modelling_26.Sep.duf	*Mine asbuilts and schedule for P2N cave.*
GEOTECHNICAL	Faults	• DXF files for each fault.	*Fault surfaces for consideration in cave back geometry.*
	Lithology	• DXF files for each lithology.	*Lithology boundaries for consideration of stability of cave back.*
	Technical Note	• P2N Hydraulic Radius.pdf	*Technical figure outlining expected critical hydraulic radius of the P2N undercut.*

	FILE TYPE	FILE NAME(S)	COMMENTS
	Technical Report	• R_01-1092-04_SRK_RTINTO_20230614_RevC.pdf	*SRK report on rock mass strength for OT.*
	Technical Memorandum	• 20210323 - 01 Geotechnical Information for OT P0 Air Blast Nomogram Development_R1.pdf	*Memorandum regarding typical airblast conditions for P0 cave – used for comparison of mechanism.*

4 ASSUMPTIONS

The following assumptions were applied to the scope:

- Airblast modelling was carried out based on the provided VentSim™ model and the understanding of the current ventilation circuit.
- The VentSim™ model provides a true and adequate reflection of the current mine network. Where the simulations will likely be impacted by changes to the mine ventilation network this has been highlighted as much as foreseeable.
- The base assumption for the modelling is the sudden failure/collapse of the critical hydraulic radius supplied by the client.

5 VENTILATION MODEL AUDIT

Prior to creating a VentSim™ Caving Simulation model, a ventilation model audit was conducted based on the framework provided within the software. Changes to parameters to complete airblast modelling are outlined below.

5.1 Airway Parameters

5.1.1 Resistance

All airway resistances are set as automatic unless specific for ventilation infrastructure.

When the model was supplied by the client, no resistances had associated strength ratings. As part of the airblast modelling, pressure ratings were estimated and included in the pre-sets (Table 5-1). The "airblast plug" resistance type with no pressure rating was added to determine peak pressures on walls during airblast modelling on a given proposed plug. This is applicable to the in-built Caving Airblast Simulation Tool only.

Table 5-1: Resistances and strength ratings for current ventilation infrastructure used for airblast modelling.

Resistance Name	Resistance Ns²/m8	Reference Density kg/m³	Strength Rating kPa	Reversing Resistance (leave zero for default) Ns²/m8
Blocked	∞	**1.2**		∞
Bin / Ore Pass Full	100000	1.2		0
Blind Closed	2	1.2		0
Blind Open	0.004	1.2		0
Blocked Drawpoint	25	1.2		0
Concrete Plug	100000	1.2		0
Door - DWS - Closed	0.015	1.2		0
Door - Roller - Closed	20	1.2	5	0
Door_PVC_Door_Closed	2	1.2	3	0
Door_Roller_Fire_Closed	20	1.2	5	0
Door_Roller_Fire_Open	0.0006	1.2		0
Door_Roller_Leaky	2	1.2	5	0
Door_Vehicle_Closed	12	1.2	3	0
Door_Vehicle_Closed_Le...	2	1.2	3	0
Door_Vehicle_Closed_V...	0.3	1.2	3	0
Door_Vehicle_Open	0.0006	1.2		0
MeshBrattice	3	1.2	5	0

It should be noted that resistance such as regulators and orifices cannot have a pressure rating associated with them and as such need to be replaced with custom resistances if necessary. When auditing the ventilation network, it was noted that auto-regulators with equivalent orifice size were used in several locations pertinent to the study (Figure 5-1). As these do not have realistic resistance values to replicate, they were first replaced with louvre type auto-regulators to match required flows then swapped out for equivalent resistances with strength ratings assigned. Additionally, conventional orifices and regulators were also replaced with equivalent resistances with strength ratings as shown in Table 5-2.

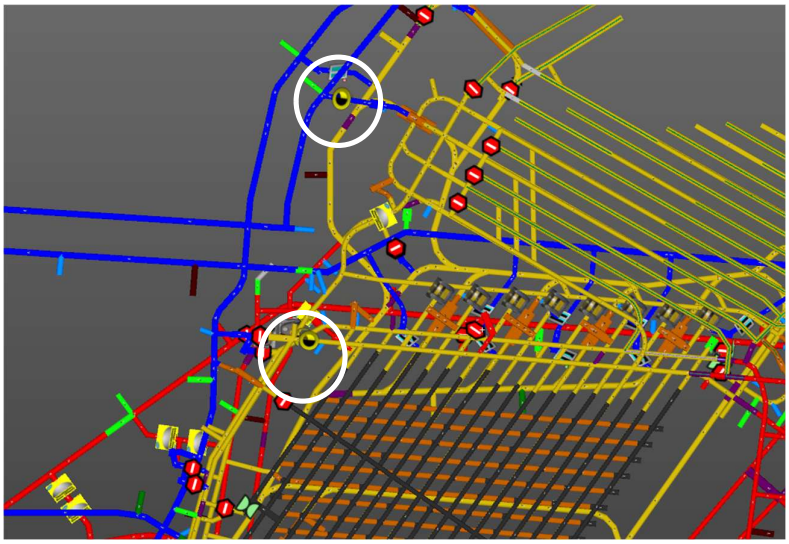

Figure 5-1: Locations of equivalent orifice type auto regulators in the client supplied model.

Table 5-2: Equivalent resistances for regulators and orifices.

Resistance Name	Resistance Ns²/m8	Reference Density kg/m³	Strength Rating kPa	Reversing Resistance (leave zero for default) Ns²/m8
SH2 - Heater Fans	1E-07	1.2		0
Vent Curtain_Closed	4	1.2		0
Wall W/PA door - Closed	500	1.2	10	0
Wall W/PA Door - Open	0.543	1.2		0
Orifice 0.1m2	155.66	1.2	10	0
Orifice 0.5m2	6.304	1.2	10	0
Orifice 4m2	0.09593	1.2	10	0
Orifice 4.7m2	0.0665	1.2	10	0
Orifice 5m2	0.06245	1.2	10	0
Regulator (Auto) 100%	0.02094	1.2	10	0
Regulator (Auto) 85.7%	0.0362	1.2	10	0
Regulator (Auto) 84.3%	0.0377	1.2	10	0
Regulator (Auto) 55.7%	0.1465	1.2	10	0
Regulator (Louvre) 100%	0.02096	1.2	10	0
Regulator (Louvre) 40%	0.2796	1.2	10	0
Regulator (Louvre) 35%	0.3894	1.2	10	0
Regulator (Louvre) 30%	0.498	1.2	10	0
Regulator (Louvre) 20%	1.123	1.2	10	0
Regulator (Louvre) 10%	4.504	1.2	10	0
Regulator (Louvre) 0%	476.68	1.2	10	0
Regulator (DBR) 0%	60.93	1.2	10	0
Airblast Wall	1000	1.2		0

It was also revealed in the audit that there are fixed resistances used throughout the model (Figure 5-2). It is unknown to RMS what these are for and therefore, they were not altered for this analysis. However, it is recommended that the client review these resistances and replace them with more representative elements in the model going forward.

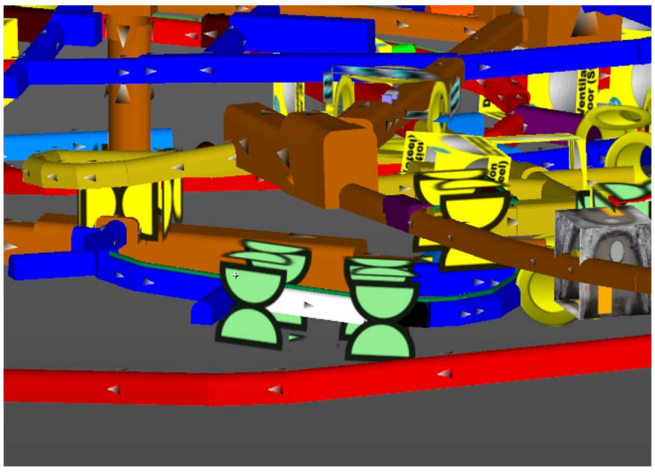

Figure 5-2: Example of fixed resistance values being used in the model.

5.1.2 Friction Factors

Friction factors in the model appear to be based on measurement and airway sizes and seem to be reasonable (Table 5-3).

Table 5-3: Friction factor presets in the model.

# in use (total all stages)	Friction Name	Friction Factor kg/m³	Roughness m	Reference Density kg/m³	Comment
1554 (4892)	**Custom**				
48 (101)	**Auto**				
194 (391)	Capital Level_shotcrete	0.01	0	1.2	
	Chinese Protan	0.002	0	1.2	Nominal K
13 (26)	Concrete Lined Airway	0.0033	0	1.2	
	Concrete Shaft Sets	0.0417	0	1.2	
4 (8)	Drive <15m2	0.014	0	1.2	
1223 (2808)	Drive 15-20m2	0.013	0	1.2	
291 (809)	Drive 20-25m2	0.012	0	1.2	
4473 (11569)	Drive 25-35m2	0.011	0	1.2	350m for drives
1548 (3513)	Drive 35m2 >	0.009	0	1.2	
92 (308)	Flexible Duct	0.0029	0	1.2	
3 (6)	RaiseBored Airway	0.005	0	1.2	
3 (6)	Raise_Unlined 1.4m Diameter	0.005	0	1.2	15M for raises
62 (141)	Raise_Unlined 2.4m - 3.0m Diameter	0.005	0	1.2	
7 (17)	Raise_Unlined 4.5m - 6.0m Diameter	0.004	0	1.2	
83 (242)	Steel Lined Raise	0.0029	0	1.2	
	Shaft #1 (MVS calculation)	0.013	0	1.2	0.01625 MVS
20 (40)	Shaft #1 (UP/DN Survey)	0.00845	0	1.2	UP/DN Survey results - 28 Feb 23
	Shaft #2 (estimate)	0.009	0	1.2	
	Shaft #2 (Check)	0.018	0	1.2	Adjusted to balance 1&2 Shaft
12 (24)	Shaft #2 (UP/DN Survey)	0.01653	0	1.2	UP/DN Survey results - 28 Feb 23
15 (30)	Shaft #3 (estimate)	0.0001	0	1.2	Adjusted after Vent survey 28 Aug
12 (43)	Shaft #4 (estimate)	0.004	0	1.2	
6 (12)	Shaft #5	0.003	0	1.2	
1 (2)	Smooth Blasted	0.0085	0	1.2	
	Timbered Airway	0.0333	0	1.2	
10 (20)	Very Rough Blasted	0.02	0	1.2	

5.1.3 Airway Size

There are several airway sizes employed in the OT Mine and all appear to be utilised appropriately (Figure 5-3).

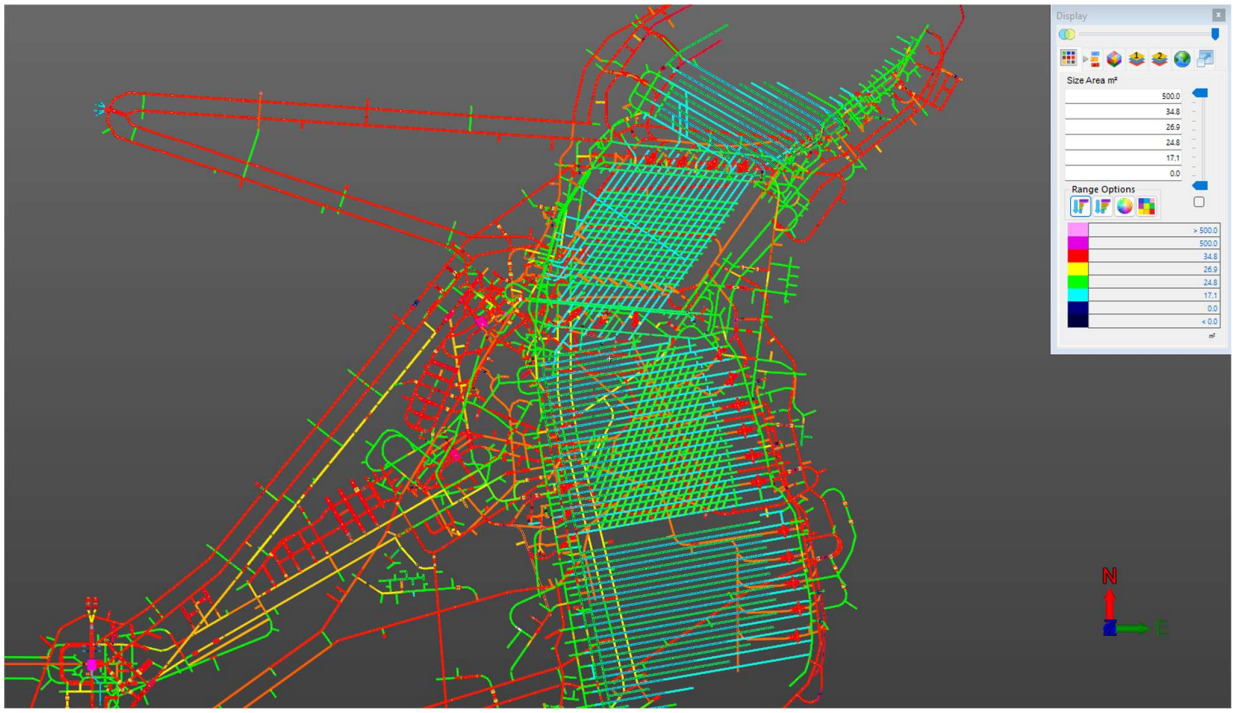

Figure 5-3: Airway size distributions of the model.

5.1.4 Airway Shape

Airway shapes are largely assigned based on development (arched or square) or raiseboring (circular) profiles. Given the level of detail in assigning profiles by mine area, no further investigation was done, and they were assumed correct.

5.1.5 Shock Factors

Some shock factors are included in the model but the application doesn't appear consistent. However, given the application of custom resistance factors, additional shock losses may not be necessary. No ventilation survey was available to consider the accuracy of the model and it was therefore assumed that the application of shock losses is not required.

5.1.6 Airway Lengths

The only fixed length airways in the model are associated with mass excavation and therefore, should not pose an issue during modelling. All other airway lengths seem appropriate in the model.

5.2 Vent Ducts

No audit was completed on the vent ducts in the model as these are immaterial to the airblast modelling.

5.3 Fans

No in-depth audit was completed on vent fans in the model as these are immaterial to the airblast modelling.

5.4 Environmental Parameters

5.4.1 Height Datum

Surface height is set at 1,173.4m above sea level with a surface datum for the mine grid being 1,160.9m. This does not have an impact on the airblast simulations conducted here.

5.4.2 Surface Barometric Pressure

Air density has been adjusted so barometric pressure is calculated based on surface height datum.

5.4.3 Surface Air Density

The surface air density is given as 1kg/m^3 which is appropriate given the model elevation.

5.4.4 Surface Temperatures

Surface temperatures are given as 32.4°C/17.3°C for dry and wet bulb respectively. As they do not factor into airblast modelling the appropriateness has not been confirmed.

5.5 Simulation Parameters

5.5.1 Fan Pressure Method

This is not relevant for airblast modelling.

5.5.2 Compressible Flows

Compressible flows were changed to “incompressible flows”. Incompressible flows are more appropriate for airblast modelling due to the fast change in volume flows and pressure which create a pressure wave ahead of the failing plug.

5.5.3 Natural Ventilation Pressure

Natural Ventilation pressure was turned off for modelling. This function should only be used if a representative heat analysis has been done but is not relevant for airblast modelling.

5.6 Thermodynamics

This is not applicable to airblast simulation and thus was not audited.

5.7 Structural Integrity

5.7.1 Airway Connections

No issues were found regarding airway connections.

5.7.2 Surface Connections

All surface connections are present.

5.7.3 Closed Areas Blocked or Excluded

Future development is either blocked or excluded via staging. Changes were made depending on the airblast scenarios that were run as described in the individual chapters.

5.7.4 Water Suspension Check

Water suspension can occur between shaft velocities of 7 m/s to 12 m/s but is not relevant for airblast modelling.

5.7.5 Stages Check

The stage "Jan28 - P2S UC Start" was used as the baseline for modelling and adjusted to include only development and other excavations scheduled to be mined at the time of cHR. Other changes to the network or further stages created, such as airblast plug deployment, are described in detail in the relevant sections of this report.

5.8 Airway Size Optimisation

This is not applicable to airblast simulation and thus was not audited.

5.9 Model Velocities

Velocities inherent to the ventilation network were checked to ensure excess speeds were verified and to provide context to the airblast results within this report. From this review, no abnormal or excessive velocities were in the model prior to airblast simulation (Figure 5-4).

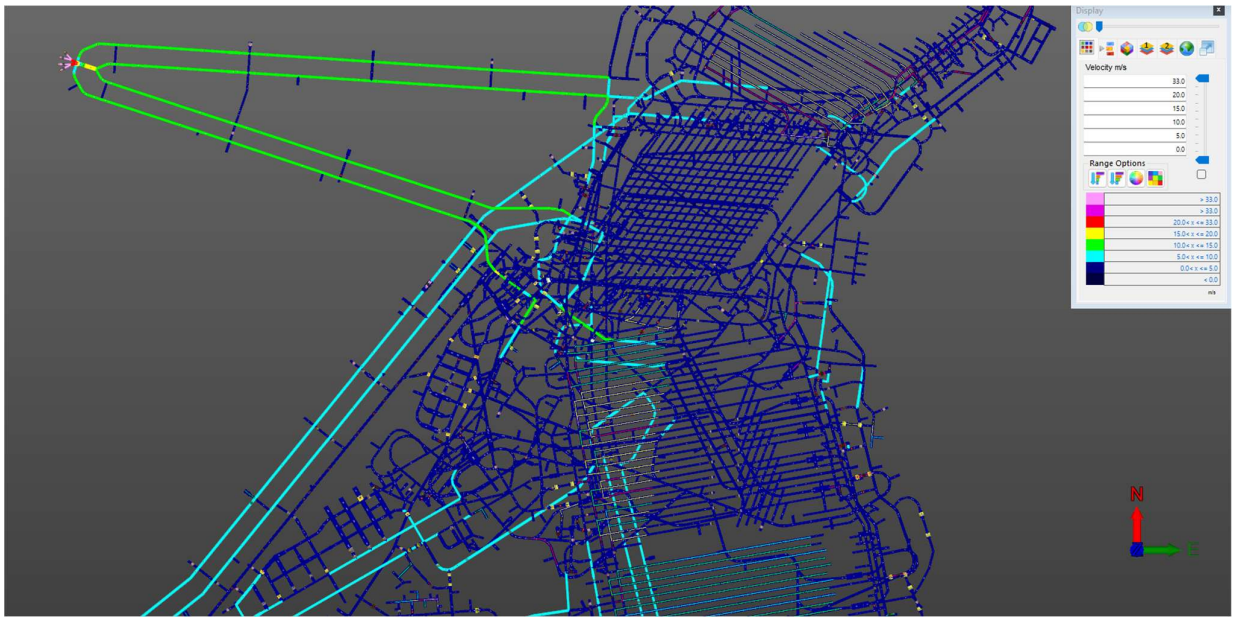

Figure 5-4: Model velocities prior to airblast simulation.

6 CAVING AIRBLAST SIMULATION TOOL (CAST) – VENTSIM™

The Caving Airblast tool available in VentSim™ models an estimation of the airblast within the mine network. The main benefit is that downstream mine resistances can be modelled dynamically as the airblast progresses. The different heights of the drives connecting into the cave are taken into consideration when determining the areas that are most affected by the airblast.

The CAST modelling is based on the following features:

- Isentropic compression of the air within the air gap of the cave utilising remaining airgap height or porosity to limit compression.
- Air velocities in the mine network are calculated at each timestep based on the pressure differentials within the mine network.
- Network pressures are based on the interaction with the cave at each time step.
- At each timestep, the full mine network is modelled to a steady state. This might lead to an overestimation of the impact extent of very small events such as stope failures but has not been found to impact sublevel and block caving models.
- All cave connections can be included in the models. Drives higher up in the cave are blocked at the time step when caving material passes the drive elevation. It is thus important that the vertical extent of the model reflects mine conditions as closely as possible.
- Airblast plug loads are determined at each timestep and represent the simulated pressure only. Design safety factors need to be applied to these loads based on the mine's risk

management framework to determine design criteria for the civil engineers.

- In general, the pressure differential created by the airblast will be used to simulate the directional air flow based on the modelling algorithms in VentSim™, which are based on thermodynamic laws and fluid dynamics.

When modelling airblasts in VentSim™ the following limitations are currently applicable:

- VentSim™ creates a circular cave model based on an input of cave diameter, muckpile height and cave back height. Rectangular cave backs can also be modelled using width and length of the cave, but use the same mechanisms as a circular cave of the same area.
- When connecting drives to the cave only z-coordinates are taken into consideration. If there are larger lateral differences, these must be accounted for manually as VentSim™ does not use northing or easting as spatial inputs.
- The lowest connected drive in the model is assumed to be the base of the muckpile. If specific real-life scenarios are run, muckpile heights potentially must be adjusted based on what the mine uses as the base of the muckpile.
- The connections to the airblast model are done in intervals based on user input. The intervals start from the lowest RL. Care should be taken when selecting interval size as this has the potential to significantly affect local air flows. However, the difference in air velocities in the main accesses will still be comparable.

The principal layout of the cave created as part of the simulation is show in Figure 6-1.

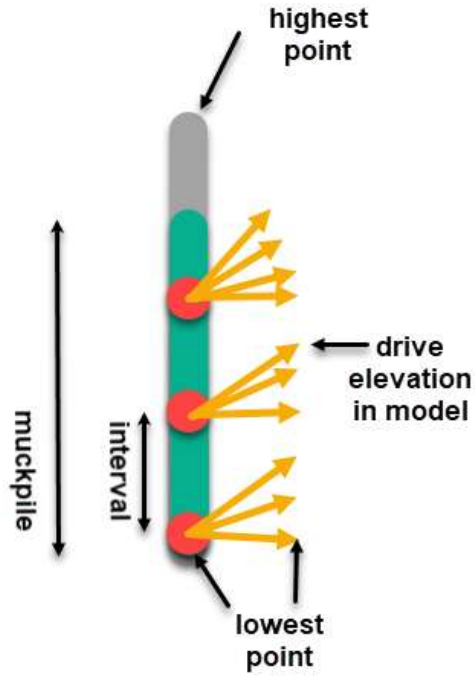

Figure 6-1: Graphical Input Parameters for Caving Airblast Model.

6.1 Swell Factors and Muckpile Resistance

CAST does not require taking swell factors into consideration. The swell factor is indirectly implied in the muckpile resistance. A smaller swell factor leads to a higher muckpile resistance and a larger swell factor leads to a lower muckpile resistance. The muckpile resistance applied generally uses a swell factor for a more mature cave (around 15 to 20 %). This is done due to the uncertainty of cave growth. The flow through the muckpile is proportional to the inverse of the square root of the resistance:

$$Q \sim \frac{1}{\sqrt{R}}$$

Where Q is the volume flow and R is the resistance. If the muckpile resistance for a more immature cave is used and cave growth stalls, then the impact of the muckpile porosity increasing due to secondary fragmentation as material is drawn down will lead to potentially hazardous conditions in the mine.

For cave simulations, Resolve Mining Solutions use a resistance for modelling in lieu of any other data of 0.0203 Ns^2m^{-8}/ vertical m / 10,000m^2, which gets adjusted based on the area undercut using Atkinson's Law:

$$R_2 = R_1 * \frac{A_1^2}{A_2^2}$$

The value is based on the back calculations performed by Andrew Derrington after the Northparkes airblast (2004) as well as measurements taken at the PC1 cave by Jako Kok (2014), which indicated that the resistance is influenced by the size of the undercut area (Vejrazka, 2016). Further measurements at another mine site on a younger cave with less porosity confirms that the initial resistance is higher, therefore using the above value is a conservative approach. OT currently uses a factor of 0.0018 Ns^2m^{-8}/ vertical m unadjusted for undercut area, which is the calculated value from Andrew Derrington's work for Lift 1 and equates to a value of 0.0203 Ns^2m^8/ vertical m /10,000m^2 when adjusted.

There is a practical lower limit to the muckpile resistance which is not covered by the above equation and research in that area is ongoing across multiple universities. The South Dakota School of Mines & Technology has published a paper based on measurements at Henderson and CFD modelling (2021) coming to the same conclusion. The practical limit for large panel caves appears to be just above 0.0006 Ns^2m^{-8}/ vertical m.

7 P2N PRE-CRITICAL HYDRAULIC RADIUS MODEL

7.1 Model Build and Scenarios

Upon review of client supplied information, the Panel 2 North cave at Oyu Tolgoi is expected to have a critical hydraulic radius for propagation of 18m or equivalent span of 72m. Due to the advance undercut methodology for cave establishment, the critical hydraulic radius of undercutting is likely to be surpassed prior to significant propagation of the rock mass. In this scenario it is likely that the cHR is effective once the material within the undercut saw tooth has been bogged allowing the wedges of material between apex pillars to fail. At this point, it is possible for the cave to form a continuous airgap which could potentially lead to an airblast risk. Therefore, for this study, the cHR was represented as the area of producing drawbells behind the undercut front which form a regular span of >72m (see Figure 7-1). Using this failure geometry, it was possible to establish the resistance characteristics for the model (Table 7-1). To account for increased muckpile across drives which are still choked with undercut material, a variable resistance profile was established in the model based on the concept outlined in Figure 7-2.

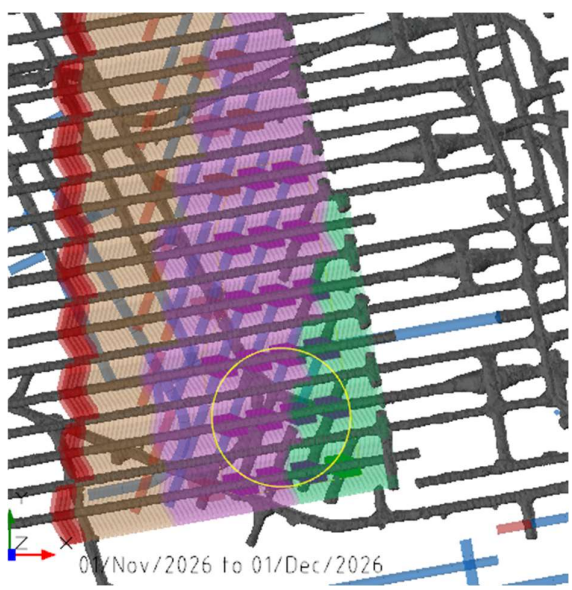

Figure 7-1: Critical hydraulic radius (yellow) shown vs. undercut at time of writing (green), at time of critical hydraulic radius (purple) and beyond (brown).

Table 7-1: CAST model inputs for cave back geometry.

Parameter	Value	Unit
Critical Span	72	m
Cave back failure area	4072	m^2
Normalised muckpile resistance	0.0203	Ns^2m^{-8}/vertical m/10,000m^2
Muckpile resistance	0.1224	Ns^2m^{-8}/vertical m
Linear resistance for Ventsim	12.24	Ns^2m^{-8}/100 m
Collapsed void minimum height	2	m

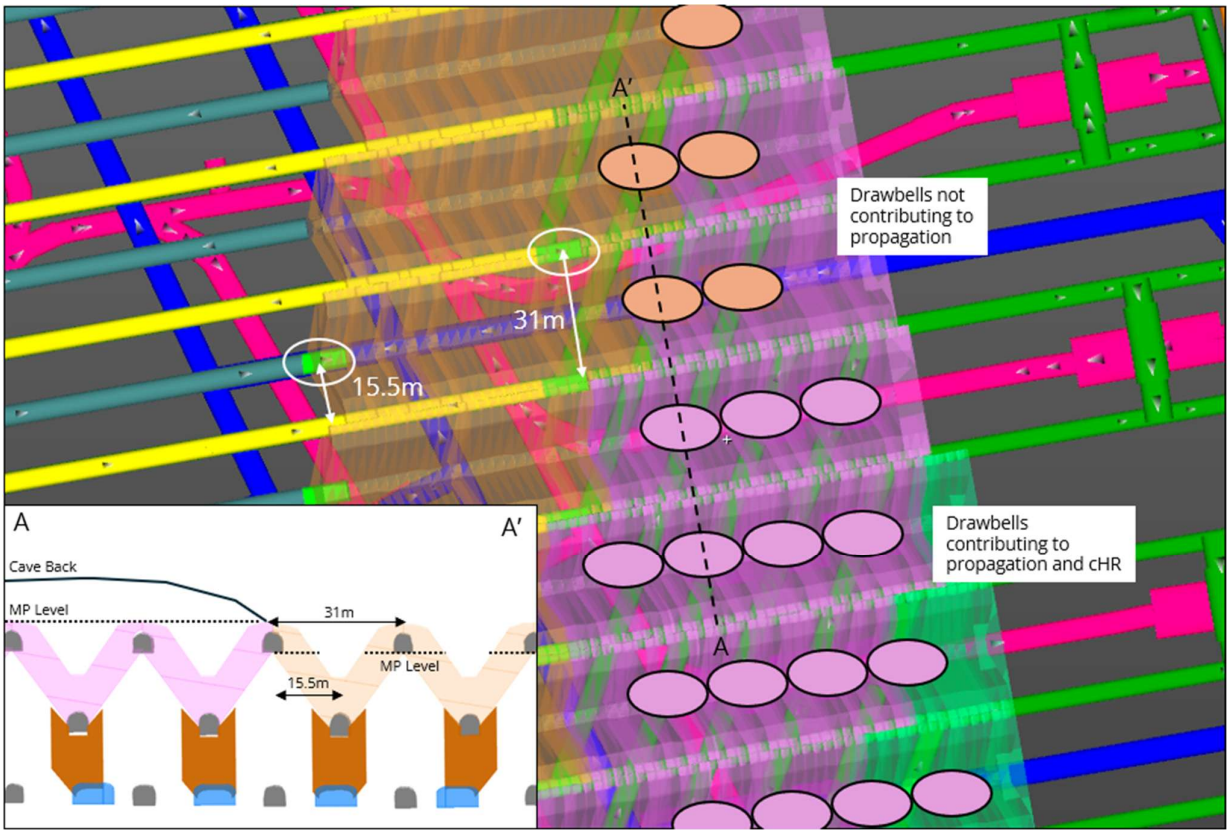

Figure 7-2: Geometry of variable muckpile resistance used on edges of cave.

The critical period for the P2N cave is during the early propagation with limited span (or spans close to the cHR). During this time, it may be possible to deplete the muckpile with little to no corresponding cave propagation which could serve to exacerbate the risk of airblast. To properly characterise the early propagation of the P2N cave, a series of scenarios were selected to cover a broad range of possible geometries. As shown in Figure 7-3, two scenario groups were selected, the open apex and choked apex sets. For the open apex scenarios, the muckpile is assumed to be either at the level of the apex drive (cracked brow) or below and still above the undercut drive. These model runs are considered worse case if production continued without growth of the cave. The second set of scenarios were setup to determine the maximum airgap where limited muck above the apex level can be considered an effective control against excessive velocities. A full detailed summary of each scenario and its geometries are given in Table 7-2.

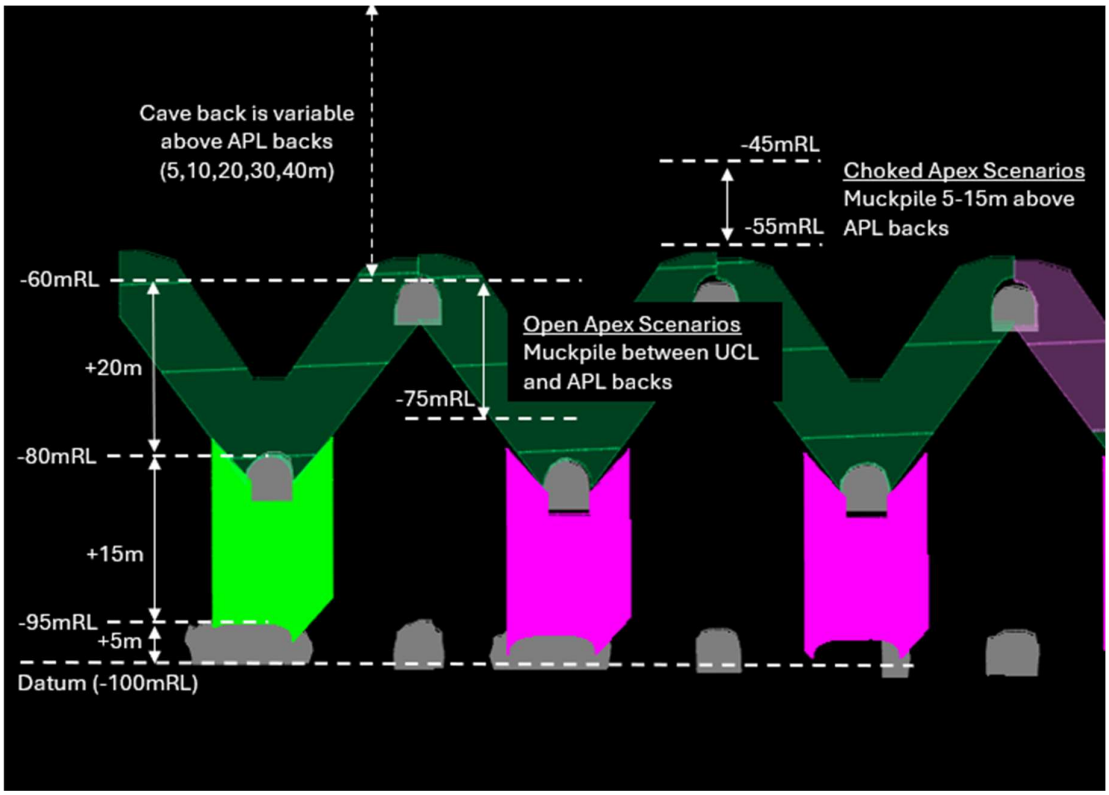

Figure 7-3: Conceptual model for critical hydraulic radius (early propagation) airblast scenarios.

Table 7-2: Early propagation airblast scenarios for P2N.

Scenario Group	Cave Back mRL	Muckpile mRL	Muckpile Height for Simulation (m)	Airgap (m)	Airgap Volume (m3)	Muckpile Above Apex Level (m)	Muckpile Above Undercut Level (m)
Open Apex Level	-55	-60	40	5	20,360	0	20
	-55	-65	35	10	40,720	Below	15
	-50	-60	40	10	40,720	0	20
	-55	-70	30	15	61,080	Below	10
	-50	-65	35	15	61,080	Below	15
	-55	-75	25	20	81,440	Below	5
	-50	-70	30	20	81,440	Below	10
	-50	-75	25	25	101,800	Below	5
Choked Apex Level	-50	-55	45	5	20,360	5	25
	-40	-45	55	5	20,360	15	35
	-40	-50	50	10	40,720	10	30
	-40	-55	45	15	61,080	5	25
	-30	-45	55	15	61,080	15	35
	-30	-50	50	20	81,440	10	30
	-30	-55	45	25	101,800	5	25
	-20	-45	55	25	101,800	15	35
	-20	-50	50	30	122,160	10	30
	-20	-55	45	35	142,520	5	25

7.2 Model Results

7.2.1 Muckpile Below Apex Level (Open Apex)

For the open apex scenarios, the velocities experienced in the apex level (APL) exceed safe levels for operation if an airblast was to occur. As early cave draw is planned from drawbells offset from the undercut front (advanced undercut method), open connection through to the apex level is considered unlikely if the bogging of swell for firing is carefully controlled. Figure 7-4 shows that if muck is in place in the apex level drive and the brow is choked, the velocity is restricted to manageable levels. However, if the muckpile is further depleted down towards the undercut level (UCL), the airblast is severe for the APL.

As the muckpile is depleted further toward the UCL, the airblast velocities are gradually reduced for the APL due to more air being siphoned off by the UCL as a result of lower muckpile resistance. However, the pathway for air through the muck into the undercut level is likely to be more restricted and these results are too heavily reliant on input assumptions for use in risk mitigation. Figure 7-5 shows the velocity profile of an open apex drive case. The air is concentrated on the apex and undercut perimeters and then expelled to the wider mine areas. The doors at the access to P2N are modelled to fail in the event of an open apex airblast.

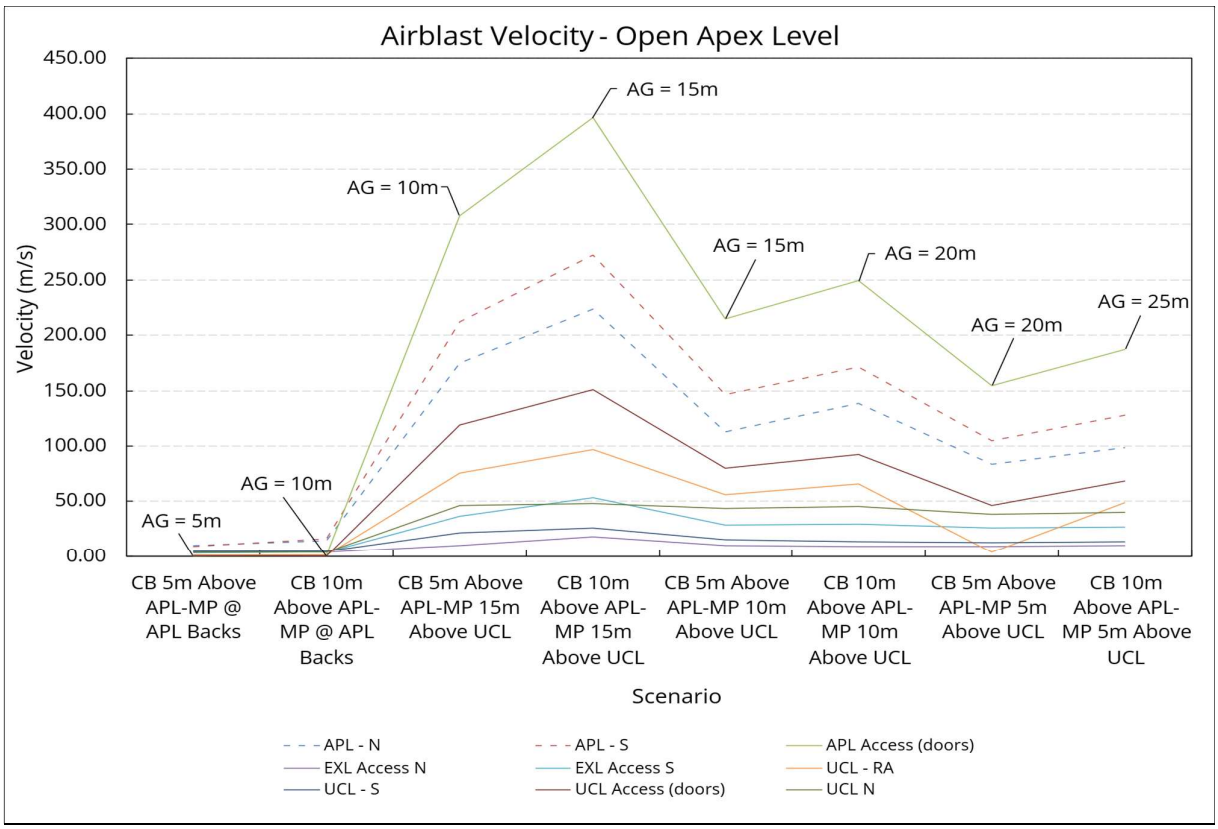

Figure 7-4: Airblast velocities for open apex scenarios (CB is cave back height above apex level APL, MP is muckpile height

above undercut level UCL and AG = airgap height).

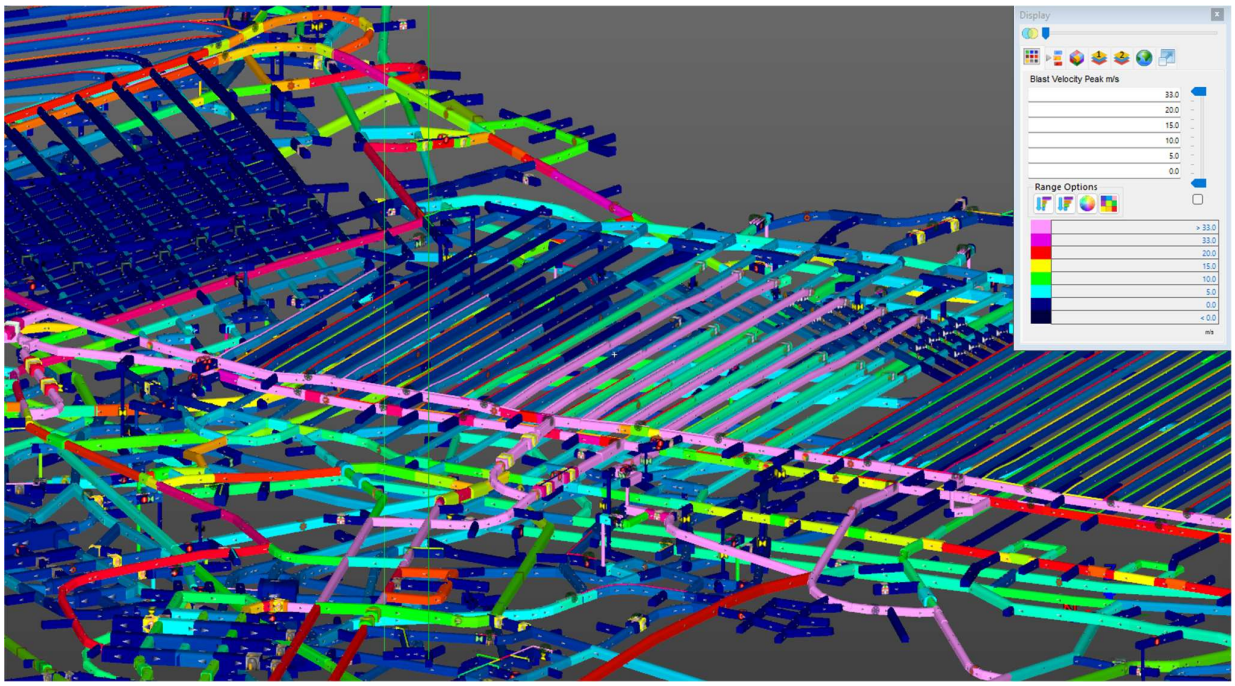

Figure 7-5: Airblast velocities for muckpile below APL and 5m above UCL (Cave back modelled at -50mRL).

It is important to note that the very high velocities displayed by the model are the result of the perfect piston and incompressible flow assumptions used for modelling. The assumptions are useful to indicate the level of hazard present for worst case conditions. However, if failure was to occur over a longer time, or as an unravelling type of failure, the volume of air injected per second would be decreased. This was calculated for the worst-case scenario by way of average volumetric flow in Table 7-3. For this calculation, incompressible flow, seven (7) connected apex level drives and a constant failure volume were assumed to provide a direct comparison to the modelled worst case. It is also important to note that the velocities experienced at the end of a failure would exceed the average rates quoted and therefore, a full risk assessment should be conducted prior to the use of any simplified calculation process.

Table 7-3: Volumetric flow rate for different cave back failure times.

Displaced Volume of Air (m3)	Time of Failure (s)	Average Flow Rate of Air into Workings (m3/s)	Average Velocity into APL Drives (m/s per drive)*
61,080	2	30,540	241
61,080	4	15,270	121
61,080	10	6,108	48
61,080	20	3,054	24
61,080	60	1,018	8

*assumes flow into 7x apex level drives.

As shown by the first principles approach, the failure time required to sufficiently decrease the air velocities to manageable levels (<20m/s) is expected to be unlikely in the event of a cave collapse. Therefore, in the case of the P2N mine, it is recommended to ensure (through design and monitoring) that there is muckpile cover for each apex level drive connected to the cave at all stages of cave propagation.

7.2.2 Muckpile Above Apex Level

When the muckpile is at levels higher than the apex level brow, the early propagation airblast hazard is greatly reduced in the model. For various combinations of muckpile and cave back heights, velocities are modelled to be either below or in line with the commonly used upper threshold of 20m/s (Figure 7-6). In general, the modelling shows that if a maximum airgap height of 25m or equivalent volume of ~80,000m^3 is coupled with muckpile cover above the apex of 5m or greater, the velocities experienced in the P2N workings are manageable through appropriate risk management and the implementation of access restrictions and TARPs.

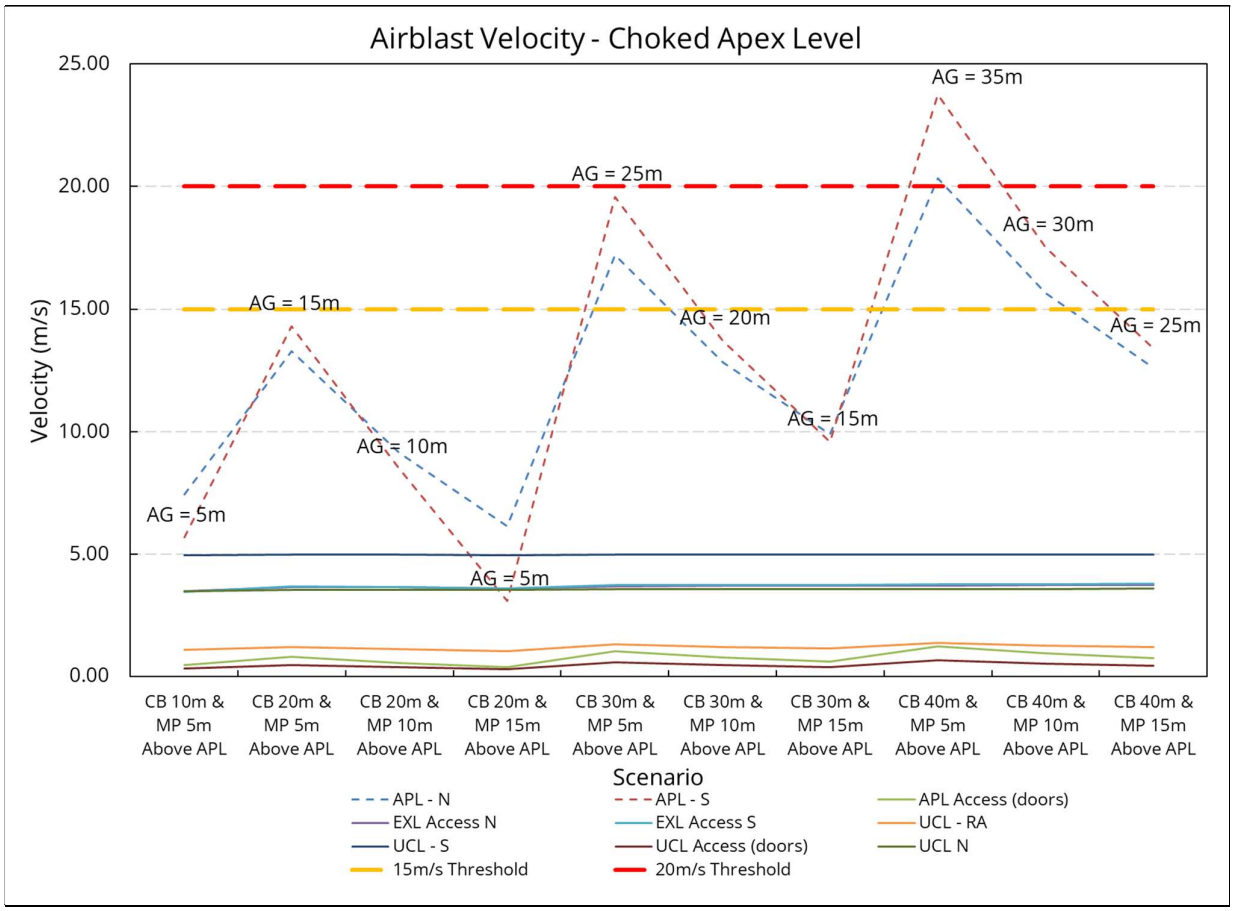

Figure 7-6: Airblast velocities for choked apex scenarios (CB is cave back height above apex level APL, MP is muckpile height

and AG = airgap height).

8 AIRBLAST CONTROLS AND MITIGATION

For adequate airblast hazard management, a full risk assessment and resulting Trigger Action Response Plan (TARP) and Airblast management plan should be implemented to document the controls required by site to mitigate the risk. However, in general, there are six (6) main methods for mitigating the risk of airblast. These are (in order of control effectiveness and hierarchy):

1. Controlled crown pillar failure (Elimination)
    a. Artificially inducing failure in the cave crown pillar to break through to surface. This eliminates the airblast risk.
2. Connections to surface (Engineering)
    a. Redirecting airblast flow to surface via existing or additional ventilation pathways (pit adits, shafts, declines and/or raises).
3. Airblast plugs (Engineering)
    a. Installation of engineered plugs to isolate high risk areas of the mine circuit.
4. Ventilation circuit changes (Engineering, Administration)
    a. Opening or closing pathways for air travel to redirect the hazard away from working areas and critical infrastructure. This control can be eroded by operational changes in ventilation circuits down the track, if airblast risk is not fully understood.
5. Void management (Administration, Engineering)
    a. To maintain as small a void as possible to minimise the air available for the airblast hazard if it were to occur.
6. Access restrictions (Administration)
    a. Removing personnel and equipment from at risk areas or levels of the mine. This can be an engineering control if hard barricades are installed preventing access.

Due to the early stage of the cave being evaluated, the controlled crown pillar failure control is not applicable to P2N. However, each of the other control types were evaluated at a high-level to determine suitability for further consideration by site.

8.1 Connection to Surface, Airblast Plugs and Ventilation Circuit Changes

The primary pathways for airblast if one was to occur is along the apex and undercut perimeters and towards the P0 and P2S caving areas. If severe, a third pathway through the failed doors at the accesses for P2N could also see excessive velocities towards the workshops and the access shaft (see Figure 8-1).

Due to the centralised location of the P2N cave with respect to the mine wide ventilation network,

changes to the circuit to divert the airblast air to surface via return air and/or the placement of airblast plugs are limited or impossible. Therefore, they are not recommended to control the airblast hazard for the P2N cave.

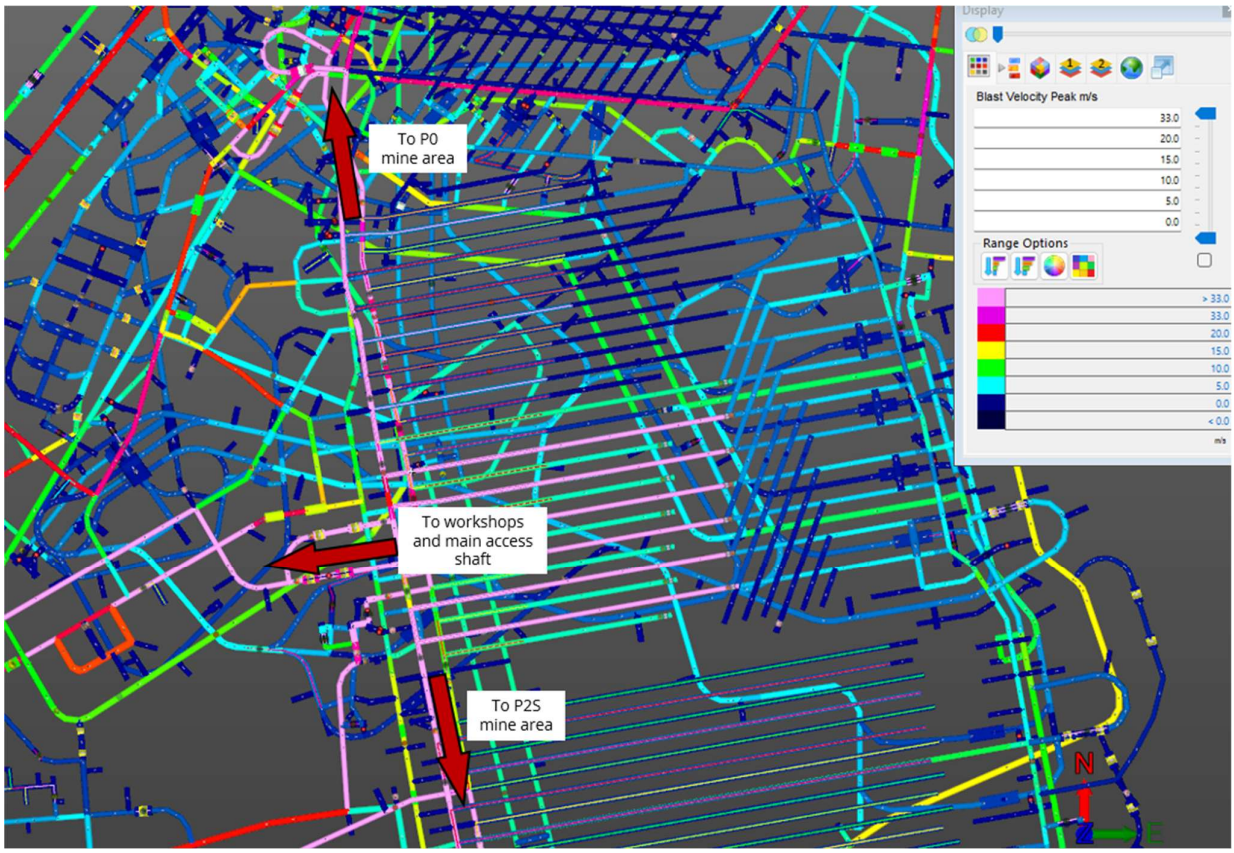

Figure 8-1: Direction of primary airblast flows.

8.2 Void Management

Potentially the most effective airblast control strategy for the P2N cave is to manage the airgap height/volume and the level of muckpile providing resistance to flow into the mine workings. As detailed in the supplied technical note (Geotechnical Information to Support OT P0 Air Blast Nomogram Development – 2021), the early propagation of the cave is likely to be represented by the conceptual geometry displayed in Figure 8-2. As such, it is critical that the maximum amount of material be in place in the leading areas of undercut that are not contributing to the producing span. It is recommended to carefully manage mass balance to ensure the muckpile is maximised at the early stages of cave propagation to provide maximum resistance to flow. The cave back and airgap monitoring should then inform production planning to ensure the volume of air available for potential compression is minimised.

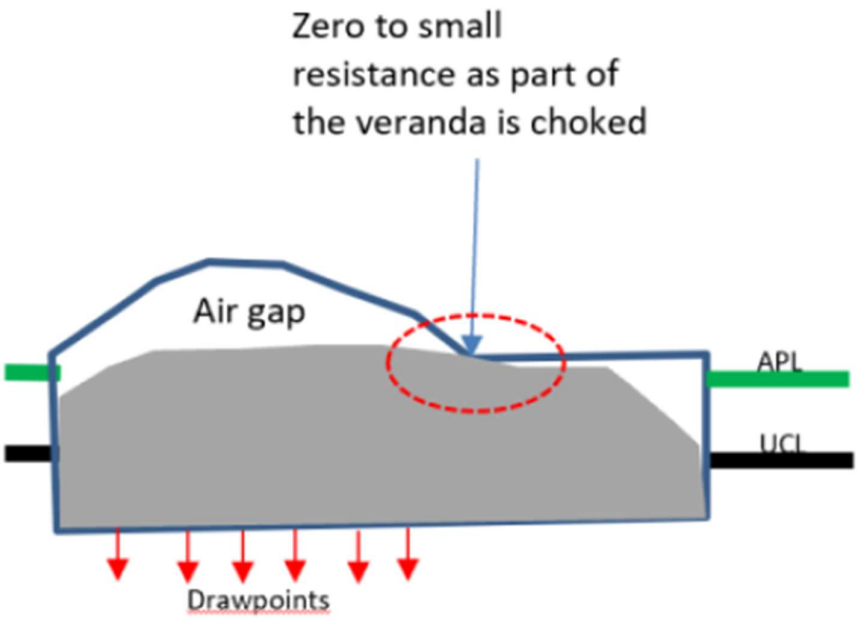

Figure 8-2: Early propagation muckpile control to avoid open apex scenarios (from Geotechnical Information to Support OT P0 Air Blast Nomogram Development – 2021).

8.3 Access Restrictions

Another secondary layer of control which can be implemented for the P2N footprint is to restrict access to the apex level during the early stages of propagation to minimise exposure to personnel. In addition, care should be taken to ensure that access restrictions are extended to areas downstream of potential airblast such as the P0 and P2S.

It is also recommended to update any risk assessment and TARP associated with airblast to cover potential muckpile heights and conditions highlighted in this report. Then via continued modelling of airblast based on real time conditions, the risk can be assessed and appropriate controls implemented prior to the hazard escalating.

9 CONCLUSION

The pre-critical Hydraulic Radius (pre-cHR) airblast modelling for the Panel 2 North (P2N) cave at Oyu Tolgoi has identified key risks associated with early cave propagation and the potential for hazardous airblast events. The modelling, conducted using the VentSim™ CAST tool, demonstrates that airblast velocities can exceed safe operational thresholds under certain cave geometries—particularly when muckpile levels are depleted below the apex level and airgaps are significant.

The study confirms that the most critical period for airblast risk is during early cave propagation when spans approach or exceed the critical hydraulic radius without sufficient muckpile resistance. In such scenarios, the velocity of airblast through apex and undercut drives could pose a serious threat to personnel and infrastructure.

While engineering controls such as airblast plugs and ventilation circuit changes are limited in effectiveness due to the centralised location of the P2N cave, administrative and operational controls, in particular, void management and access restrictions offer viable mitigation strategies. The modelling supports the conclusion that maintaining muckpile levels and minimising airgap volumes are essential to reducing airblast risk.

10 RECOMMENDATIONS

Resolve Mining Solutions recommends the following actions to mitigate airblast risk in the P2N cave:

1. Implement Robust Void Management Practices
    a. Maintain high muckpile levels in undercut areas not contributing to cave propagation ensuring a significant veranda of muckpile material is available between the active cave void and personnel work areas in the undercut.
    b. Monitor cave back and airgap volumes in real time via direct means wherever possible or via mass balance to inform production planning and minimise air gap volumes.
2. Establish Access Restrictions and Update TARPs
    a. Restrict access to the Apex Level and downstream areas during early cave propagation. Access should be linked to valid available controls (for example seismic monitoring if applicable).
    b. Update site-specific Trigger Action Response Plans (TARPs) to reflect the findings of this modelling, including thresholds for muckpile heights and airgap volumes.
3. Conduct Ongoing Airblast Modelling
    a. Continue to model airblast scenarios using updated mine conditions to assess evolving risks. Ensure that changes to the ventilation circuit trigger an airblast model update.
    b. Use real-time data to refine control strategies and ensure proactive hazard management.
4. Review and Strengthen Risk Assessments
    a. Incorporate the findings of this report into site-wide risk assessments.
    b. Ensure that airblast hazard is appropriately classified and managed within the site's risk framework.

5. Engage in Cross-Functional Collaboration
    a. Ensure coordination between geotechnical, ventilation, and operations teams to maintain alignment on cave development and hazard mitigation strategies.

11 REFERENCES

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